Lecture 6:
QUANTUM MEASUREMENT THEORY

- *Much of the material in this chapter is a review of concepts already introduced. Nevertheless, if this is your first exposure to quantum theory, it is important that you master these topics.*
- *Measurement plays a fundamental role in quantum computation because, at some point, we have to be able to get information out the computational system. In this chapter we will learn the basics about different measurement models used in quantum theory.*

Lecture 6: QUANTUM MEASUREMENT THEORY

When the quantum physical observables are measured,

- *what measurement results are possible and what the probability is of obtaining each measurement result.*
- *what the state of the system is after a measurement is made. While measurement generally has no effect on a system in classical mechanics (i.e. macroscopic system in general),*
- *Measurement has a profound impact on a quantum mechanical system—altering its state in an irreversible way.*

DISTINGUISHING QUANTUM STATES AND MEASUREMENT

An action of measurement disturbs a quantum system in a fundamental way.

For example, a qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

When a measurement is made, the qubit will be forced into the state $|\psi\rangle \longmapsto |0\rangle$ *or* $|\psi\rangle \longmapsto |1\rangle$. *After measurement the original state is lost.*

It isn't possible to make a measurement and determine what α *and* β *are.*

Review 1:

- *A quantum system is described by a vector (the "state vector") in a Hilbert space. The state of the system at time t is denoted by $|\psi(t)\rangle$.*
- *The dynamical behave of a quantum system is determined by the Hamiltonian operator H, which describes the total energy of the system.*
- *The time evolution of an isolated or closed system is described by the Schrödinger equation $i\hbar\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle$, its general solution gives us the state of the system at time t. So*

$$|\psi(t)\rangle = e^{-iHt/\hbar}|\psi(0)\rangle$$

Review 2:

- *The Hamiltonian* H *is Hermitian, the* $e^{-iHt/\hbar}$ *is a unitary operator, so the unitary evolution operator:*

$$U = e^{-iHt/\hbar}$$

- *If the system is initially described by some density operator* ρ_0*, then the state of the system at time* t *will be*

$$\rho_t = U\rho_0 U^\dagger$$

- *The dynamics of a quantum system is trace-preserving. This means that if the system is initially described by some density operator* ρ_0 *with* $Tr(\rho_0) = 1$*, then after the system has evolved to a final state describe by* ρ_t*, then* $Tr(\rho_t) = 1$*.*

- *While time evolution is trace-preserving, measurement is described by trace-decreasing quantum operation. A quantum operation involving measurement, described by a measurement operator that we will denote by* M_m*, transforms a density operator* ρ *according to* $\rho' = M_m \rho M_m^\dagger$*. In this case,* $Tr(\rho') \leq 1$*.*

PROJECTIVE MEASUREMENTS(*Von Neumann)*

- *The easiest to understand,*
- *It is how introductory quantum mechanics is usually taught,*
- *It's the oldest type of measurement model.*
- *It plays a fundamental role in quantum information theory.*

Given a set of mutually exclusive possible states, what state is the system in?

For example:

1. *An atom could have two mutually exclusive states—a lower energy "ground state" denoted by* $|g\rangle$ *and an "excited state" denoted by* $|e\rangle$*.*
2. *The position of a particle. Is it located at position* x_1 *or at position* x_2*?*
3. *Is the qubit* $|0\rangle$*, or is it* $|1\rangle$*?*

A projection operator P *is Hermitian*

$$P = P^{\dagger} \text{ and } P^2 = P$$

Two projection operators P_1 *and* P_2 *are orthogonal if their product is zero. That is, for every state* $|\psi\rangle$,P_1 *and* P_2 *are orthogonal if*

$$P_1 P_2 |\psi\rangle = 0$$

- *A set of mutually exclusive measurement results corresponds to a set of orthogonal projection operators that act on the state space of the system.*
- *A complete set of orthogonal projection operators is one for which*

$$\sum_i P_i = I$$

- *Every complete set of orthogonal projectors specifies a measurement that can be realized.*
- *If a set of orthogonal projection operators is complete, implies that at least one of the possible measurement results must be true.*

  –the reason that probabilities must sum to one.
- *The number of projection operators is determined by the dimension of the Hilbert space that describes the system. If the dimension of the Hilbert space is* d *and there are* m *projection operators, it must be true that*

$$m \leq d$$

For example:

Given a qubit where $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$*, the dimension of the space is 2, and the projection operators corresponding to the mutually exclusive measurement results* $|0\rangle$ *and* $|1\rangle$ *are*

$$P_0 = |0\rangle\langle 0|, \qquad P_1 = |1\rangle\langle 1|$$

$$P_0 \times P_1 = ?$$

The projection operators corresponding to the ground and excited states of the atom would be written as

$$P_g = |g\rangle\langle g|, \qquad P_e = |e\rangle\langle e|$$

- *If two projection operators commute, then their product $P_1 P_2$ is also a projection operator however, the sum of two or more projection operators is, in general, not a projector.*
- *The necessary and sufficient condition needed for the sum of a set of projection operators to be a projection operator is that they be mutually orthogonal.*
- *If there is a set of projection operator $\{P_1, P_2, P_3, \ldots\}$, we can indicate that they are mutually orthogonal by writing*

$$P_i P_j = \delta_{ij} P_i$$

Let the dimension of the system be n*, and consider a set of mutually orthogonal projection operators* $\{P_1, P_2, P_3, \dots, P_n\}$*. The system is prepared in a state* $|\psi\rangle$:

The probability of finding the ith outcome when a measurement is made is

$$\Pr(i) = |P_i|\psi\rangle|^2 = (P_i|\psi\rangle)^\dagger(P_i|\psi\rangle) = \langle\psi|P_i^2|\psi\rangle = \langle\psi|P_i|\psi\rangle$$

Some observable set of projection operators

We denote it by A, and let the eigenvectors of A be denoted by $|u_i\rangle$ each with an eigenvalue a_i. The spectral decomposition of A allows us to write the operator as

$$A = \sum_{i=1}^{n} a_i |u_i\rangle\langle u_i| = \sum_{i=1}^{n} a_i P_i$$

Where the projection operator corresponding to measurement outcome a_i is given by $P_i = |u_i\rangle\langle u_i|$.

We can expand the state of the normalized system $|\psi\rangle$ in terms of the eigenvectors of A as

$$|\psi\rangle = \sum_{i=1}^{n} (\langle u_i|\psi\rangle)|u_i\rangle = \sum_{i=1}^{n} c_i |u_i\rangle$$

Here $c_i = \langle u_i|\psi\rangle$ is the probability amplitude for obtaining measurement result a_i when the system is in the state $|\psi\rangle$. The actual probability for the given measurement result is found by computing by Born rule--- $\Pr(i) = |\langle u_i|\psi\rangle|^2$.

Some observable set of projection operators

If the eigenvalue is degenerate, then the probability is found by summing over all eigenvectors that correspond to that eigenvalue

$$\Pr(i) = \sum_j \left|\langle u_j|\psi\rangle\right|^2$$

Recalling that the trace turs outer products into inner products (i.e., $Tr\left(A\left|\psi\right\rangle\langle\psi|\right) = \langle\psi|A|\psi\rangle$*), notice that the probability of obtaining measurement result* a_i *can be written as*

$$\Pr(i) = \langle\psi|P_i|\psi\rangle = Tr(P_i|\psi\rangle\langle\psi|)$$

The state of the system after measurement.

Collapse of the wave function: What this means is that while the state of the system prior to measurement could be a superposition of basis states, after measurement the system collapses to the basis state that corresponds to the measurement result that was obtained.

Formally, we write the state of the system after measurement $|\psi'\rangle$ *as*

$$|\psi'\rangle = \frac{P_i|\psi\rangle}{\sqrt{\langle\psi|P_i|\psi\rangle}}$$

Where the factor $\langle\psi|P_i|\psi\rangle$ *in the denominator is to ensure that* $|\psi'\rangle$ *is normalized.*

- *The expectation value or average of an observable A with respect to a state $|\psi\rangle$ is given by*

$$\langle A\rangle = \sum_i a_i \langle\psi|P_i|\psi\rangle$$

For a system with a single qubit, we can write the orthogonal projection operator as

$$P_\pm = \frac{I \pm \vec{n}\cdot\vec{\sigma}}{2}$$

$$P_\pm = \frac{I \pm \hat{x}\cdot\vec{\sigma}}{2} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1| \pm |0\rangle\langle 1| \pm |1\rangle\langle 0|)$$

,

Example 6.1

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \frac{\sqrt{6}}{\sqrt{19}}|u_5\rangle$$

Where $|u_i\rangle$ $(i = 1 - 5)$ *form a complete and orthonormal set of vectors. Each* $|u_i\rangle$ *is a eigenstate of the system`s Hamiltonian corresponding to the possible measurement result* $H|u_i\rangle = n\varepsilon|u_i\rangle$, *where* $n = 1,2,3,4,5$.

- *Describe the set of projection operators corresponding to the possible measurement results.*
- *Determine the probability of obtaining each measurement result. What is th state of the system ofter measurement if we measure the energy to be* 3ε*?*
- *What is the average energy of the system?*

Example 6.2

A qubit in the state

$$|\psi\rangle = \frac{\sqrt{3}|0\rangle - |1\rangle}{2}$$

A measurement with respect to Y is made, given that the eigenvalue of the Y matrix are ± 1*, determine the probability that the measurement result is* $+1$ *and the probability that the measurement result is* -1*.*

Example 6.3

A system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{6}}|0\rangle + \sqrt{\frac{5}{6}}|1\rangle$$

A measurement is made with respect to the observable X. What is the expectation o average value?

MEASUREMENTS ON COMPOSITE SYSTEMS

- *Composite systems were introduced in Chapter 4. Many applications in quantum computation involve composite systems. In this section we will go over some basic measurement operations on composite systems.*

Example 6.4

Describe the action of the operators $P_0 \otimes I \; and \; I \otimes P_0$ *on the state*

$$|\psi\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Example 6.5

A system is in the state

$$|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

What is the probability that measurement finds the system in the state $|\psi\rangle = |01\rangle$*?*

What is the probability that measurement finds the first qubit in the state $|0\rangle$*?*
What is the state of the system after measurement?

Example 6.6

A three-qubit system is in the state

$$|\psi\rangle = \frac{\sqrt{2}+i}{\sqrt{20}}|000\rangle + \frac{1}{\sqrt{2}}|001\rangle + \frac{1}{\sqrt{10}}|011\rangle + \frac{i}{2}|111\rangle$$

a) *Is the state normalized? What is the probability that the system is found in the state* $|000\rangle$ *if all 3 qubits are measured?*

b) *What is the probability that a measurement on the first qubit only give 0? What is the postmeasurement state of the system?*

Example 6.7

A system is in the state GHZ where

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

suppose that an observable $A = \sigma_x \otimes \sigma_y \otimes \sigma_z$, *if* $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ *what is the probability that measurement finds the system in the state* $|+++\rangle$ *and in the state* $|---\rangle$ *when the system is in the state* $A|\psi\rangle$*? What is the expectation value of* A*?*

Example 6.8

A qubit system is in the state

$$|\phi\rangle = \frac{\sqrt{3}}{2}|00\rangle + \frac{1}{2}|11\rangle$$

A Y gate is applied to the first qubit. After this is done, what are the possible measurement results if both qubits are measured, and what are the respective probabilities of each measurement results?

Measurement operators satisfy a completeness relation that follows from the fact that probabilities sum to one.

For general measurement operators

$$\sum_m M_m^\dagger M_m = I$$

After a measurement the state of the system is

$$|\psi'\rangle = \frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}$$

. *If a quantum system is described by a density operator* ρ*, the probability of finding measurement result* m *is*

$$\Pr(m) = Tr\left(M_m^\dagger M_m \rho\right)$$

If the measurement in question is described by a set of orthogonal projection operators $P_i = |u_i\rangle\langle u_i|$ *corresponding to measurement result* i *then the probability of finding that measurement result is*

$$\Pr(i) = Tr\left(P_i^\dagger P_i \rho\right) = Tr(|u_i\rangle\langle u_i|\rho) = \langle u_i|\rho|u_i\rangle$$

The state of a system described by a density operator after obtaining measurement result m *is given by*

$$\rho' = \frac{M_m \rho M_m^\dagger}{Tr\left(M_m^\dagger M_m \rho\right)}$$

In the case where the measurements are described by a set of orthogonal projection operators and measurement result i *is obtained, the state of the system after measurement is*

$$\rho' = \frac{|u_i\rangle\langle u_i|\rho|u_i\rangle\langle u_i|}{\langle u_i|\rho|u_i\rangle}$$

Example 6.9

A quantum system has a density matrix given by

$$\rho = \frac{5}{6}|0\rangle\langle 0| + \frac{1}{6}|1\rangle\langle 1|$$

What is the probability that the system is in the state $|0\rangle$*?*

Example 6.10

A system has a density operator

$$\rho = \frac{1}{3}|u_1\rangle\langle u_1| - i\frac{\sqrt{2}}{3}|u_1\rangle\langle u_2| + i\frac{\sqrt{2}}{3}|u_2\rangle\langle u_1| + \frac{2}{3}|u_2\rangle\langle u_2|$$

Where the $|u_k\rangle$ *constitute an orthonormal basis. What is the propbability that a measurement finds in the* $|u_2\rangle$*?*

POSITIVE OPERATOR-VALUED MEASURES(POVM)

More general than the projective measurements;

A POVM consists of a set of positive operators commonly denoted by E_m*. The probability of obtaining measurement result* m *in this case is given by*

$$\Pr(m) = \langle\psi|E_m|\psi\rangle$$

When the system is a mixed state described by a density operator ρ*, the probability of obtaining measurement result* m *is given by* $Tr(E_m\rho)$*.*

In addition to being positive operators, the set of E_m *satisfy*

$$\sum_m E_m = I$$

The measurement operators in a POVM can be constructed from an arbitrary measurement operator by taking

$$E_m = M_m^\dagger M_m$$

It is possible to form a POVM to describe a projective measurement, but the operators E_m $do\ not\ have\ to\ be$ *projection operators.*

The POVM allow us to construct a more general type of measurement operator to describe measurements where projective measurements do not apply in the real world.

For example, if a system is in a state $|\psi\rangle = \sum_i c_i |u_i\rangle$*, a projective measurement* $|u_k\rangle\langle u_k|$ *collapses the wave function to the state* $|u_k\rangle$ *. We are free to repeat another measurement immediately on the system—and that measurement will find the system in the state* $|u_k\rangle$ *with certainty.*

In the laboratory not all measurements are repeatable.

The quintessential example is the detection of a photon—after it has been detected, the photon is destroyed. Hence repeated measurements on the system are not possible. A POVM is applicable in this case because it allows us to describe measurements on the system without regard to the post measurement state.

POVM's can be useful when projective measurements are not. For example, POVM's provide the ability to distinguish between non-orthogonal states.

Example 6.11(describe measurements on the system without regard to the post measurement state.)

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{5}}|0\rangle + \frac{1}{\sqrt{5}}|1\rangle$$

Describe the probabilities of measuring 0 and 1 for this state in the POVM formalism.

Example 6.12 (The possibility of imperfectly distinguishing between nonorthogonal states)

A system can be in one of two states $|\psi\rangle$ or $|\phi\rangle$. The states sre not orthogonal; In fact $\langle\psi|\phi\rangle = \cos\theta$. Describe the POVM that can distinguish between the two states. Assume that tha states are normalized.

Example 6.13 (Possibility of making weak measurement on a system that give us some information without causing the collapse of the wavefunction)

A POVM can be used to obtain information about a state with a weak measurement. This is a measurement that provides some information about the state while only mildly disturbing it, without forcing a "Collapse of the wave function" describe a POVM that will do this for a single qubit.